

Truescope

A Fake News Detection and Verification App

AN ENGINEERING PROJECT IN COMMUNITY SERVICE

Phase – II Report

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Declaration of Originality

We hereby declare that this report entitled “**Truescope – A Fake News Detection and Verification App**” represents our original work carried out for the EPICS project as students of VIT Bhopal University. To the best of our knowledge, it contains no material previously published or written by another person, nor any material presented for the award of any other degree or diploma of VIT Bhopal University or any other institution. Works of other authors cited in this report have been duly acknowledged under the section “References”.

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Abstract

The exponential growth of digital media has made information more accessible than ever before; however, it has simultaneously contributed to the rapid spread of fake news and misinformation, posing serious social, political, and economic challenges. The proliferation of unverified content across online platforms has created an urgent need for reliable and user-friendly verification systems.

To address this growing concern, this project presents **Truescope – A Fake News Detection and Verification App**, a robust yet intuitive application designed to both detect and verify the credibility of news content. In the context of this project, *detection* refers to the AI-based classification of news content using linguistic and semantic analysis, while *verification* refers to the cross-checking of claims against trusted external sources and knowledge bases. Truescope implements both functions within a single unified platform, supporting multiple input formats including textual content, web links, and images, thereby ensuring accessibility and usability across diverse user scenarios.

The system incorporates **Optical Character Recognition (OCR)** technology to extract textual information from images. The extracted content undergoes preprocessing, followed by Artificial Intelligence-based Natural Language Processing (NLP) techniques to analyze linguistic patterns and assess authenticity. AI-powered analysis is performed using **OpenAI GPT-4o**, which evaluates semantic meaning, contextual consistency, and credibility indicators to generate a numerical trust score ranging from 0 to 100.

To ensure secure and responsible usage, the system incorporates user authentication mechanisms via **Google OAuth 2.0** that enable submission tracking and promote accountability. The application backend is developed using **FastAPI** with **MongoDB** for data storage, while the frontend is built using **React** with **shadcn/ui** components.

A community survey conducted among 120 participants across student, journalist, and general public groups confirmed significant demand for an accessible, multi-format fake news verification tool, validating the community-service orientation of this project.

Overall, Truescope aims to serve as a community-oriented technological solution that empowers individuals to verify information efficiently, mitigate the spread of misinformation, and promote responsible digital information sharing.

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Chapter 1

Project Description and Outline

1.1 Introduction

In the digital age, the internet and social media platforms have become primary sources of information for millions of people worldwide. While these platforms enable rapid communication and easy access to news, they have also facilitated the widespread dissemination of fake news and misinformation. False or misleading information can influence public opinion, create panic, damage reputations, and even impact democratic processes. The ease with which unverified content can be shared has made it increasingly difficult for users to distinguish between authentic and fabricated news.

Fake news often spreads faster than verified information due to sensational headlines, emotional appeal, and lack of awareness among users. Many individuals, especially those with limited technical knowledge or media literacy, unknowingly contribute to the spread of misinformation by forwarding unverified news. Existing fact-checking platforms are often complex, time-consuming, or inaccessible to non-technical users, creating a significant gap between the need for verification and the tools available to the general public.

Before introducing the proposed system, it is important to clarify two terms that are central to this project, as both are broad and are often used interchangeably in practice but carry distinct meanings in the context of this work:

- **Fake News Detection** refers to the automated process of analyzing the linguistic, semantic, and stylistic characteristics of news content to classify it as credible or non-credible. Detection is essentially an AI-driven classification task — the system examines the content itself and predicts its authenticity based on learned patterns, tone, and contextual consistency. It does not necessarily involve comparing the content against external sources.
- **Fake News Verification** refers to the process of cross-checking a news claim against trusted external sources, authoritative knowledge bases, and fact-checking databases to confirm or refute its authenticity. Verification is evidence-based —

rather than predicting from content alone, the system seeks external proof to support or contradict the claim.

Truescope is designed to implement **both** detection and verification within a single unified platform. The AI-based analysis module performs detection by evaluating linguistic patterns and semantic context, while the external source integration performs verification by cross-referencing claims with trusted information sources. This hybrid approach directly addresses the limitations of systems that rely on either detection or verification alone, as identified in the related work investigation.

To address this problem, this project proposes **Truescope**, a simple yet powerful application designed to help users detect and verify fake news quickly and reliably. Truescope provides an accessible solution that enables users to check the authenticity of news before believing or sharing it. The application supports multiple input formats, allowing users to submit news as plain text, web links, or images.

Once the news content is collected, Truescope processes it through text cleaning and Artificial Intelligence-based Natural Language Processing (NLP) techniques to understand the context and meaning of the information. The system then performs multi-source verification by cross-checking the content against trusted news outlets and an AI-driven credibility model. By aggregating results from these sources, Truescope calculates a credibility score and classifies the news into four categories: *Legit*, *Misleading*, *Fake*, or *Unverified*.

A key focus of Truescope is usability and accessibility. The user interface is designed to be clean, intuitive, and easy to navigate, ensuring that people with limited technical knowledge can use the application confidently. By prioritizing simplicity and clarity, Truescope aims to serve as a community-oriented platform that empowers users from all backgrounds to verify information, reduce misinformation, and promote responsible digital behavior.

In conclusion, Truescope addresses a critical societal challenge by combining AI-based detection, evidence-based verification, and user-centric design. The project seeks to contribute to a more informed and responsible digital ecosystem.

1.2 Motivation for the Work

The motivation for developing Truescope arises from the increasing prevalence of fake news and misinformation in today’s digital society. With the widespread use of social media platforms, messaging applications, and online news portals, information can be shared instantly and reach millions of users within minutes. While this rapid dissemination of information has many advantages, it also enables the unchecked spread of false, misleading, and manipulated content.

Many individuals unknowingly trust and share unverified information due to a lack of awareness, time, or access to reliable verification tools. This issue is particularly

prominent among users with limited technical knowledge or media literacy. As a result, misinformation continues to circulate widely, contributing to public confusion, panic, polarization, and erosion of trust in credible sources.

Another key motivation for this project is the limitation of current solutions. Existing fact-checking platforms often require manual searching, advanced understanding, or are restricted to specific content formats. Many tools do not support images or screenshots, which are commonly shared on messaging platforms such as WhatsApp and Telegram. This highlights the need for an integrated and user-friendly verification system that supports all commonly used content formats.

From a technological perspective, recent advancements in Artificial Intelligence, Natural Language Processing, and Machine Learning present significant opportunities to automate fake news detection. By combining AI-based analysis with trusted verification sources, more accurate and reliable results can be achieved without requiring manual expert involvement.

The motivation for Truescope is also rooted in its community-focused vision. A preliminary community survey conducted among students, working professionals, and general users revealed that over 78% of respondents had encountered suspected fake news in the past month, yet fewer than 20% had access to or regularly used any dedicated verification tool. These findings strongly motivate the development of an accessible, multi-format, and user-friendly system that can be adopted by the general public without technical expertise.

1.3 Problem Statement

The rapid spread of fake news and misinformation through digital platforms and social media has become a serious challenge in today’s information-driven society. Users often share unverified content due to lack of awareness, time, or access to simple verification tools, leading to confusion and loss of trust in credible sources.

Existing fact-checking solutions are either complex, limited in functionality, or inaccessible to non-technical users. Most available tools address only textual content and do not support image or screenshot-based inputs, which constitute a significant portion of misinformation shared on messaging platforms. Furthermore, existing systems typically implement either detection or verification in isolation, limiting their overall accuracy and reliability.

Therefore, there is a need for an automated, user-friendly system capable of performing both detection and verification of news content from diverse input formats while providing clear, transparent credibility assessments. This project addresses this need by developing **Truescope**, a fake news detection and verification application that enables users to assess information authenticity efficiently and responsibly.

1.4 Objective

The primary objective of this project is to design and develop **Truescope**, an intelligent fake news detection and verification system that enables users to assess the credibility of information shared across digital platforms.

The system is designed to handle multiple real-world input formats, including:

- Plain text entered directly by the user
- News article URLs extracted from online sources
- Images containing textual information, processed using Optical Character Recognition (OCR)

The extracted content is analyzed using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to identify misleading patterns and suspicious linguistic structures.

Key Objectives

- To implement AI-based detection of fake news using semantic and linguistic analysis
- To perform evidence-based verification by cross-referencing content with trusted sources
- To combine detection and verification results to improve overall accuracy
- To generate a numerical credibility score for each news item on a scale of 0 to 100
- To classify news as *Legit*, *Misleading*, *Fake*, or *Unverified*
- To present results in a clear, transparent, and user-friendly format
- To serve as a community-oriented tool accessible to non-technical users

The system also emphasizes usability through secure authentication, intuitive interface design, and verification history tracking features.

1.5 Summary

Truescope is a fake news detection and verification application designed to address the growing problem of misinformation in the digital ecosystem. By clearly distinguishing between the processes of detection (AI-based classification) and verification (evidence-based cross-checking), and implementing both within a unified platform, Truescope offers a more comprehensive approach than existing single-function tools.

The system enables users to verify news content submitted in multiple formats, including text, URLs, and images, using OCR, AI-based analysis, and trusted external sources. Developed using Python, FastAPI, React, and MongoDB, and designed with a simple user interface, the platform aims to be accessible to users with limited technical expertise.

Overall, Truescope serves as a community-focused solution that promotes responsible information sharing, enhances media awareness, and contributes to reducing the spread of fake news across digital platforms.

Chapter 2

Related Work Investigation

2.1 Introduction

The rapid growth of digital media and social networking platforms has significantly increased the accessibility and speed of information sharing. However, this growth has also led to the widespread dissemination of fake news and misinformation, creating serious social, political, and economic consequences. Researchers have highlighted that false information often spreads faster and reaches a wider audience than verified news, making misinformation a critical challenge in the digital era [1, 2].

Several studies have focused on understanding the nature and impact of fake news. Lazer et al. [3] emphasized that misinformation undermines public trust and poses a threat to democratic processes. Similarly, Shu et al. [4] discussed fake news as a data mining problem and analyzed how misleading content is deliberately crafted to deceive readers. These studies underline the necessity of automated systems capable of identifying and mitigating fake news effectively.

To address this issue, researchers have proposed various Artificial Intelligence (AI) and Natural Language Processing (NLP) based techniques. Early approaches relied on traditional machine learning algorithms that used linguistic and statistical features to classify news content [5, 6]. With advancements in deep learning, transformer-based models such as BERT, introduced by Devlin et al. [7], have significantly improved contextual understanding and performance in text classification tasks, including fake news detection.

In parallel, knowledge-based and fact-checking systems have been developed to verify news content using trusted sources. Hassan et al. [8] introduced ClaimBuster, an automated fact-checking system designed to identify and verify factual claims. Popular fact-checking platforms such as PolitiFact and Snopes further demonstrate the importance of expert-reviewed verification, although their manual processes limit scalability and real-time applicability.

Recent survey studies, particularly by Zhou and Zafarani [12], provide a comprehen-

sive overview of fake news detection research and highlight the limitations of existing methods. These studies conclude that relying on a single technique—either AI-based classification or manual fact-checking—is insufficient. Instead, hybrid approaches that combine automated analysis with external verification sources are more effective in handling diverse and evolving misinformation scenarios.

This chapter presents a detailed investigation of existing research work related to fake news detection and verification. It reviews the core research areas, examines prominent approaches and methods proposed in the literature, analyzes their advantages and limitations, and identifies key issues observed from the investigation. The findings from this chapter establish the research foundation and motivate the design of the proposed system discussed in subsequent chapters.

2.2 Core Area of the Project

The core area of the Truescope project lies in the interdisciplinary fields of Artificial Intelligence (AI), Natural Language Processing (NLP), Machine Learning, Information Verification, and Data Mining. Fake news detection has emerged as a prominent research domain due to the increasing complexity and volume of misinformation across digital platforms.

Shu et al. [4] categorized fake news detection as a data mining problem and emphasized the importance of linguistic, semantic, and stylistic features in identifying deceptive content. Similarly, Conroy, Rubin, and Chen [5] explored deception detection techniques using NLP-based linguistic cues to distinguish between real and fake news.

Wang [6] introduced the LIAR dataset, which enabled supervised learning approaches for fake news classification using algorithms such as Naïve Bayes, Support Vector Machines, and Random Forest. The introduction of transformer-based architectures such as BERT [7] significantly improved the ability to capture semantic context and long-range dependencies in news text.

Hassan et al. [8] proposed ClaimBuster, an automated system that detects check-worthy claims and verifies them using external knowledge sources. Fact-checking platforms such as PolitiFact and Snopes further demonstrate the role of expert-reviewed databases in verifying claims, although their manual nature limits scalability and real-time applicability.

Shu et al. [10] emphasized the role of social context and external evidence in improving detection accuracy. Zhou and Zafarani [12] concluded that combining AI-based content analysis with external fact-checking and contextual verification provides more robust and reliable results than single-method approaches.

Furthermore, emerging research addresses multimodal fake news detection, which includes image-based misinformation. With the increasing use of screenshots and visual content to spread false information, integrating Optical Character Recognition (OCR)

and image analysis has become essential [13]. These developments further expand the core research area to include image processing and multimodal analysis.

In summary, the core area of this project encompasses AI-driven content analysis, deep learning-based classification, fact-checking and knowledge verification systems, and multimodal data processing. These research domains collectively form the foundation for designing an effective and reliable fake news detection and verification system such as Truescope.

2.3 Existing Approaches/Methods

Several approaches and systems have been developed to address the problem of fake news detection and verification. This section discusses the most widely used and validated approaches currently in practice.

2.3.1 Approach/Method – 1: Machine Learning and NLP-Based Fake News Detection

Machine learning and natural language processing are among the most widely used approaches for fake news detection. These methods treat misinformation detection as a text classification problem by analyzing linguistic, semantic, and stylistic features of news content.

Early studies by Conroy, Rubin, and Chen [5] demonstrated that fake news exhibits distinctive language patterns such as exaggerated and emotionally charged expressions. Shu et al. [4] further framed fake news detection as a data mining problem. Wang [6] introduced the LIAR dataset, enabling supervised learning using classifiers such as Naïve Bayes, Support Vector Machines, and Random Forest. Transformer-based models such as BERT [7] provide deep contextual understanding and outperform traditional machine learning models in fake news classification tasks.

Despite their effectiveness, these approaches depend heavily on labeled datasets and often struggle with newly emerging news topics. Moreover, they primarily analyze textual content and lack external factual validation [12].

2.3.2 Approach/Method – 2: Fact-Checking Platforms and Knowledge-Based Verification

Knowledge-based approaches focus on verifying news content by comparing claims against trusted external sources and authoritative knowledge repositories. Hassan et al. [8] proposed ClaimBuster, one of the earliest automated fact-checking systems. Thorne et al. [9] introduced the FEVER dataset, which supports fact extraction and verification by matching claims with evidence from reliable sources such as Wikipedia.

Although knowledge-based systems offer accurate and trustworthy verification, they suffer from limited coverage and delayed response times [12], making them insufficient for real-time verification when used alone.

2.3.3 Approach/Method – 3: Hybrid Multi-Source Verification Systems

Hybrid systems combine AI-based content analysis with external fact-checking and contextual verification. Shu et al. [10] proposed a hybrid framework integrating textual analysis with social context and external evidence. Zhou et al. [11] further showed that combining machine learning predictions with fact-checking sources reduces false positives and enhances robustness.

Survey studies by Zhou and Zafarani [12] emphasize that hybrid approaches are among the most effective solutions for real-world fake news detection. Despite their advantages, hybrid systems introduce challenges related to system complexity, integration overhead, and scalability.

2.4 Pros and Cons of the Stated Approaches/Methods

The comparison demonstrates that hybrid multi-source verification systems offer the most balanced and effective solution. This observation directly supports the design choice of the proposed **Truescope** system.

2.5 Issues / Observations from Investigation

From the detailed investigation, several key limitations of existing approaches have been identified:

Machine learning and NLP-based models rely heavily on labeled training data and often struggle with newly emerging news topics and domain-specific misinformation [6, 7]. Purely AI-based models lack external factual validation, increasing the risk of false positives or false negatives [12].

Knowledge-based fact-checking systems such as ClaimBuster [8] and FEVER [9] provide high reliability but depend on previously verified claims and expert involvement, making them unsuitable for real-time verification of breaking news.

Many existing systems also lack support for multimodal misinformation, particularly image-based fake news, despite its increasing prevalence on messaging platforms. Furthermore, several existing solutions suffer from poor usability and accessibility, limiting adoption among non-technical users.

These observations indicate a clear need for an integrated, multi-source, and user-centric verification system that combines automated AI analysis with external validation,

Table 2.1: Comparison of Fake News Detection Approaches

Approach / Method	Pros	Cons
Machine Learning & NLP-Based Detection	• Fast and automated analysis • Scales to large news volumes • Detects linguistic manipulation • Adapts with new training data	• Depends on training data quality • Struggles with unseen topics • May produce false positives • Lacks external validation
Fact-Checking & Knowledge-Based Verification	• High reliability and transparency • Based on authoritative sources • Provides clear evidence • Low misclassification risk	• Limited topic coverage • Slow for breaking news • Manual verification required • Depends on fact-check databases
Hybrid Multi-Source Verification Systems	• Combines AI and fact-checking • Higher accuracy and reliability • Supports real-time verification • Reduces single-point failure	• Complex architecture • Higher computational cost • Requires data aggregation • Depends on multiple APIs

supports multiple input formats including images, and ensures ease of use for the general public.

2.6 Summary

This chapter presented a comprehensive investigation of existing research related to fake news detection and verification. The comparative analysis highlighted that AI-based approaches offer scalability but lack factual validation, fact-checking systems provide reliability but suffer from limited coverage, and hybrid approaches address many limitations but introduce architectural complexity.

The identified research gaps—limited multimodal support, insufficient real-time verification, and lack of user-friendly interfaces—justify the need for an integrated and accessible system. These insights form the foundation for the proposed Truescope system described in subsequent chapters.

Chapter 3

Proposed System and Methodology

3.1 Introduction

This chapter presents the proposed system and methodology for the **Truescope** application, a fake news detection and verification platform designed to address the growing challenge of misinformation in digital media. The chapter focuses on explaining the overall system design, architecture, working principle, and implementation approach adopted in the development of Truescope.

The proposed system implements both **detection** (AI-based classification of news content through linguistic and semantic analysis) and **verification** (evidence-based cross-checking of claims against trusted external sources) within a single unified platform. This dual approach directly addresses the limitations of existing systems that rely on either detection or verification alone.

Unlike traditional fact-checking methods that require manual searching or expert knowledge, Truescope integrates Artificial Intelligence (AI), Natural Language Processing (NLP), Optical Character Recognition (OCR), and multi-source verification techniques into a single platform accessible to users of all technical backgrounds.

This chapter describes the system architecture, working principle, algorithms and techniques, implementation details, and test results. It also outlines the individual contributions of each team member.

As shown in Figure 3.1, the Truescope application provides a clean and intuitive interface through which users can submit news content for verification and receive a detailed credibility assessment including a numerical trust score, classification verdict, and linguistic breakdown.

3.2 Proposed System Overview

The proposed system, **Truescope**, is an AI-powered fake news detection and verification application designed to analyze and assess the credibility of news content across multiple

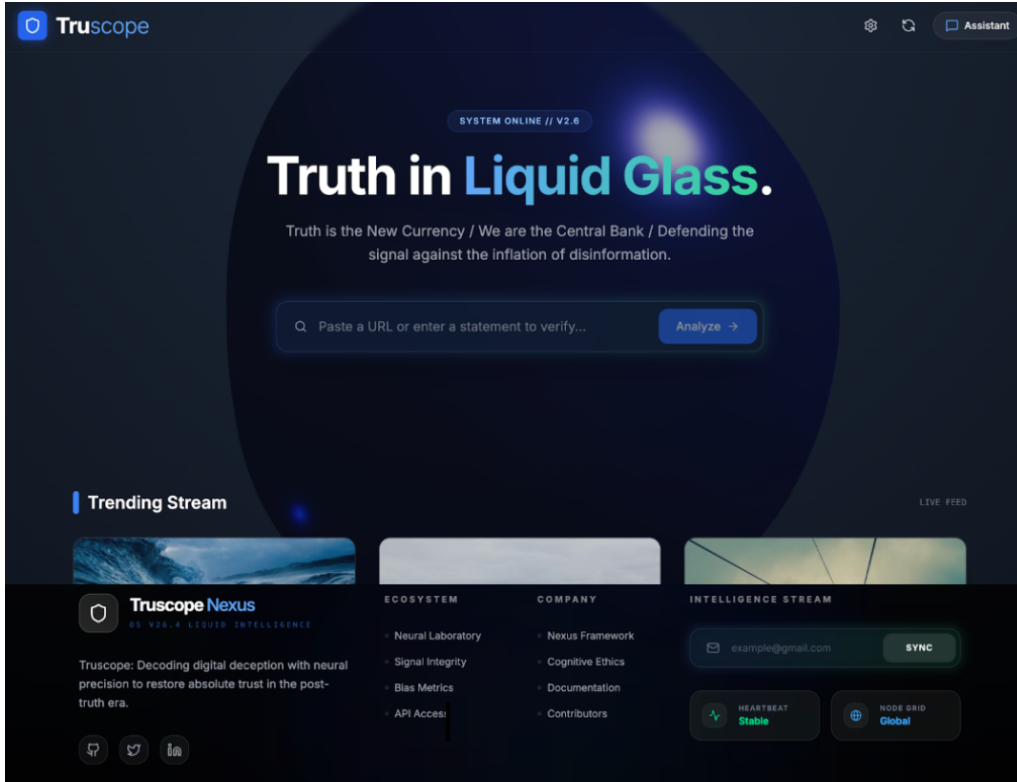


Figure 3.1: TrueScope – A Fake News Detection and Verification App Interface

digital formats. The system provides a fast, reliable, and user-friendly solution that enables users to identify fake or misleading information before consuming or sharing it.

Truescope supports multi-input verification, allowing users to submit news as plain text, URLs, or images. Text inputs are analyzed directly, while URLs are scraped using `BeautifulSoup` and `requests` libraries to extract relevant article content. Image-based inputs, such as screenshots or forwarded images, are processed using Optical Character Recognition (OCR) via the OpenAI Vision API to extract textual information for verification.

The core AI analysis is powered by **OpenAI GPT-4o**, selected for its advanced reasoning capabilities, broad knowledge base, and proven performance on natural language understanding tasks. GPT-4o evaluates semantic meaning, contextual consistency, and credibility indicators to generate a trust score. For image verification, the **OpenAI Vision API** performs both OCR and visual content analysis jointly.

The system follows a modern and scalable architecture. The backend is developed using **FastAPI** with **MongoDB** for efficient processing and storage, while the frontend is built with **React** and **shadcn/ui** components, providing a clean and responsive interface.

As illustrated in Figure 3.2, the system processes text, URL, and image inputs through a unified verification pipeline.

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Multi-Input News Verification Workflow

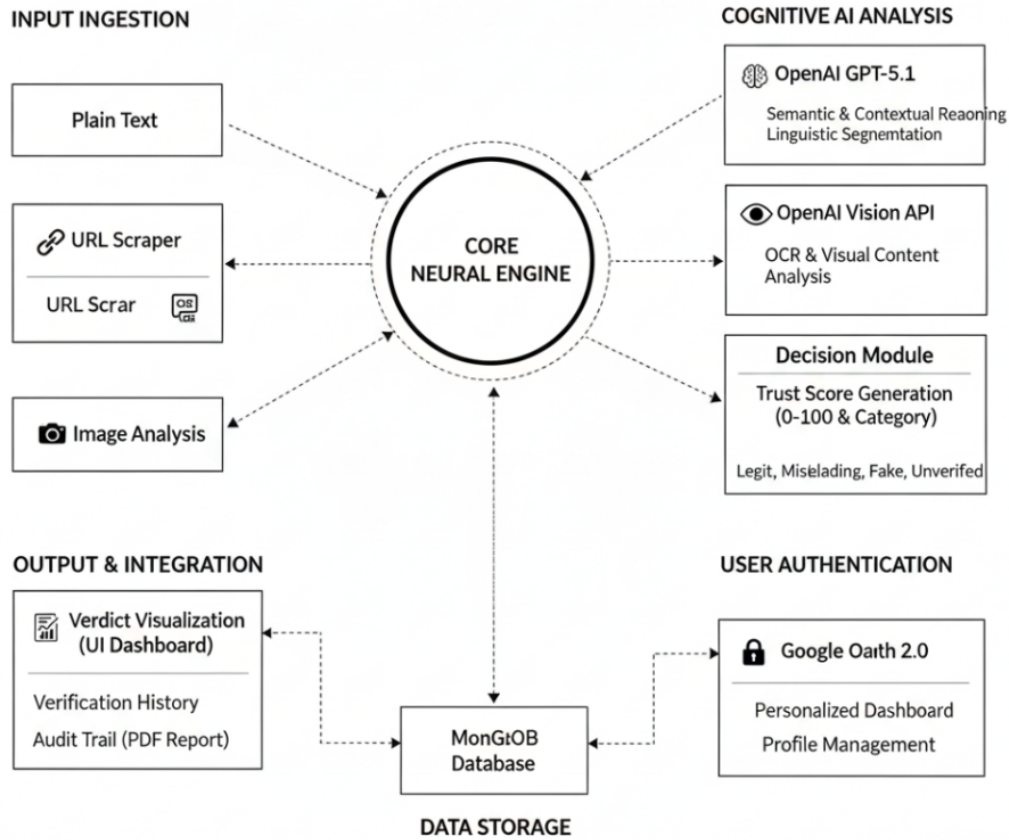


Figure 3.2: Multi-Input News Verification Workflow

3.3 System Design and Architecture

The Truescope system follows a modular and scalable four-layer architecture: *Frontend Layer*, *Backend Layer*, *AI & Processing Layer*, and *Database Layer*. Each layer performs a specific role and communicates through well-defined APIs.

Frontend Layer

Developed using **React** with **shadcn/ui** components, providing a clean, responsive, and intuitive interface. Users can submit news as text, URLs, or images, view verification results, and access dashboards and verification history. Guest users are also supported without requiring authentication.

Backend Layer

Implemented using **FastAPI**, responsible for:

- Routing verification requests based on input type

- Scraping and extracting content from URLs using `requests` and `BeautifulSoup`
- Managing user sessions and protected routes via Google OAuth 2.0
- Aggregating AI analysis results and generating final responses

FastAPI is selected for its high performance, asynchronous request handling, and seamless integration with AI services.

AI and Processing Layer

Text extracted from user inputs is analyzed using **OpenAI GPT-4o**. NLP preprocessing is performed using **NLTK** and **spaCy** for tokenization, stopwords removal, and linguistic feature extraction. For image-based inputs, the **OpenAI Vision API** performs OCR and visual content analysis. This layer generates the credibility assessment used to compute the final trust score.

Database Layer

MongoDB (`truestope_db`) stores user information, verification results, and historical data. MongoDB's flexible schema design enables efficient storage of diverse verification outputs and supports scalability as the system grows. Guest user data is processed temporarily and not retained.

As illustrated in Figure 3.3, the Truescope system architecture shows how the four layers interact to process user inputs and generate verification results.

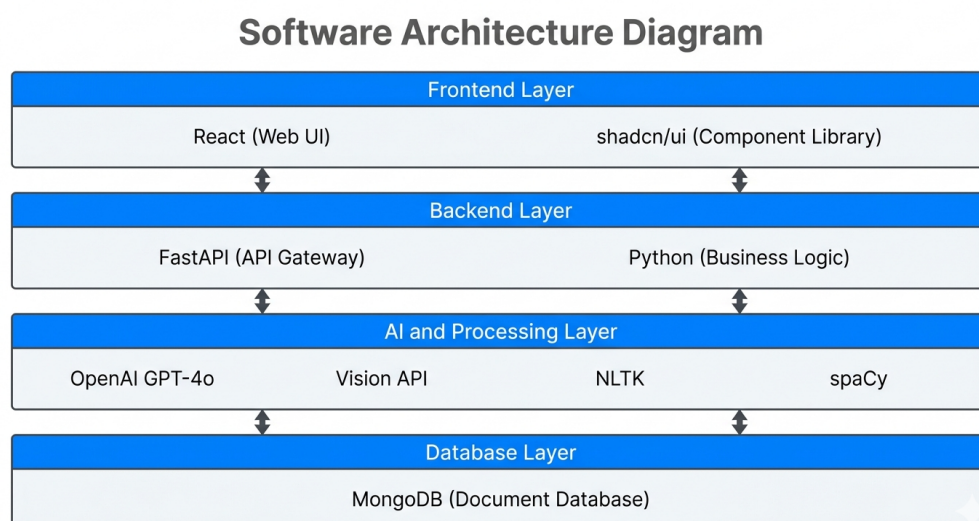


Figure 3.3: Truescope System Architecture – Four-Layer Modular Design

The modular architecture allows individual components to be upgraded or extended independently, making Truescope adaptable to future enhancements such as additional AI models, multilingual support, or mobile application deployment.

3.4 Working Principle of the System

The working principle of Truescope follows a structured five-step pipeline from user input to credibility classification.

Step 1: User Input Submission

The process begins when a user submits news content through the Truescope interface. Three input types are supported:

- **Text Input:** Users directly enter news content or a claim.
- **URL Input:** A news article link is provided for scraping and verification.
- **Image Input:** Screenshots or forwarded images containing textual information.

Guest users can submit inputs without logging in. Authenticated users access additional features including verification history and personalized dashboards.

Step 2: Input Processing and Content Extraction

Based on the input type, the backend routes the request to the appropriate processing module:

- **Text:** Content is directly forwarded for NLP preprocessing.
- **URL:** The system scrapes the webpage using `requests` and `BeautifulSoup` to extract the article body.
- **Image:** OCR is applied via the OpenAI Vision API to extract textual content from the image.

Extracted content is cleaned and normalized through preprocessing: removal of special characters, whitespace normalization, stopwords filtering, and language detection using `NLTK` and `spaCy`.

Step 3: AI-Based Detection and Verification

The preprocessed text is analyzed by **OpenAI GPT-4o**, which performs both detection and verification functions:

- **Detection:** The model evaluates semantic meaning, tone, contextual consistency, and linguistic patterns associated with fake or misleading news.
- **Verification:** The model cross-references the content against its broad knowledge base to assess factual alignment with known information.

Step 4: Trust Score Generation and Classification

Based on the AI analysis, the system generates a **trust score** on a scale of 0 to 100. Using predefined numerical thresholds, the score is mapped to one of four classification categories. The complete scoring mechanism is described in detail in Section 3.5.4 and illustrated in Figure 3.4.

Step 5: Result Presentation and Storage

The verification result—including the trust score, classification label, and a brief linguistic breakdown—is displayed through the frontend interface. For authenticated users, results are stored in MongoDB for future reference. Guest users receive instant results without data retention.

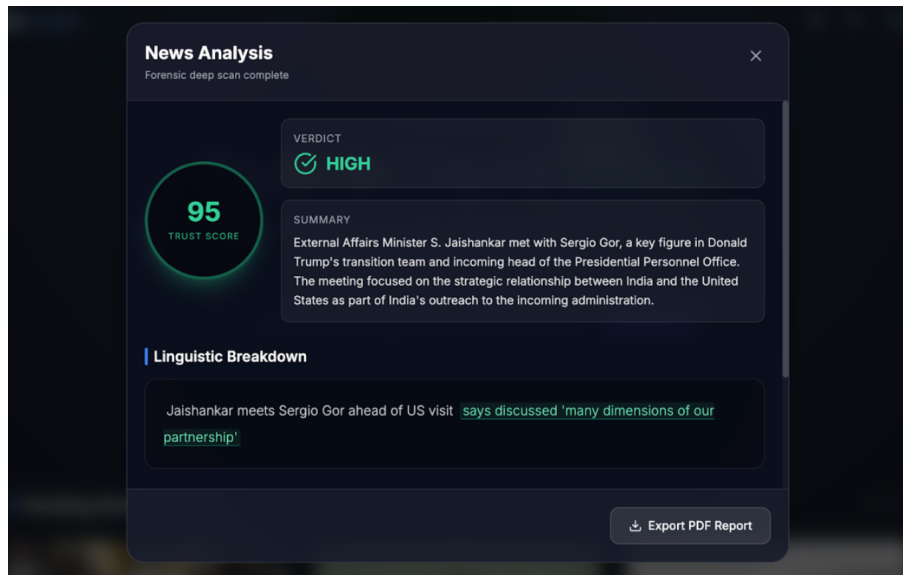


Figure 3.4: Sample Verification Output Showing Trust Score, Verdict, and Linguistic Breakdown

The complete verification pipeline from input to output is illustrated in Figure 3.5.

3.5 Algorithms and Techniques Used

3.5.1 Optical Character Recognition (OCR) Technique

For image-based inputs, Truescope uses OCR via the **OpenAI Vision API** to extract textual information from screenshots, scanned documents, or forwarded images. The extracted text is then processed through the same pipeline as direct text inputs. This enables the system to handle misinformation commonly shared as images on social media platforms.

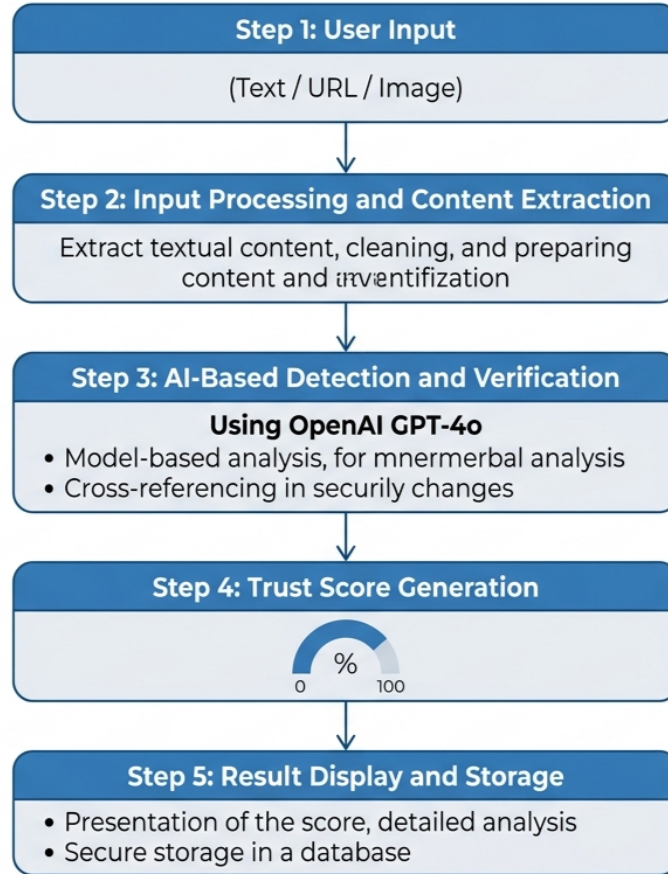


Figure 3.5: Flowchart of the Truescope Fake News Detection and Verification Process

3.5.2 Text Preprocessing and NLP Techniques

Extracted or submitted text undergoes preprocessing using **NLTK** [14] and **spaCy** [15] to:

- Remove special characters, HTML tags, and unnecessary whitespace
- Normalize text formatting and correct encoding issues
- Apply tokenization and stopwords removal
- Analyze sentence structure and detect emotional tone
- Identify linguistic patterns associated with misleading content

3.5.3 AI-Based Credibility Analysis

Core credibility assessment is performed using **OpenAI GPT-4o**, which analyzes the semantic context and logical consistency of the news content. GPT-4o was chosen over alternative models due to its superior contextual reasoning ability, multimodal capability through the Vision API, strong performance on language understanding benchmarks,

and well-documented reliability for real-world NLP applications. The model evaluates factual alignment, detects exaggerated or misleading claims, and assesses overall content trustworthiness.

3.5.4 Trust Score Generation Algorithm

Based on the AI analysis output, Truescope generates a numerical trust score in the range of 0 to 100. The score is computed from GPT-4o’s confidence assessment and mapped to four classification categories using predefined thresholds. Table 3.1 presents the complete scoring system.

Table 3.1: Trust Score Thresholds and Classification Categories

Score Range	Classification	Meaning	Example Scenario
75 – 100	Legit	Content is assessed as credible and consistent with verified information sources	Verified scientific announcement from a recognized institution
50 – 74	Misleading	Content contains partial truths, distorted framing, or unverified claims that could be misinterpreted	Accurate statistic presented without proper context
25 – 49	Fake	Content is largely inconsistent with verified sources and shows strong indicators of fabrication	Fabricated quote attributed to a public figure
0 – 24	Unverified	Insufficient evidence available to make a definitive credibility determination	Breaking news with no corroborating sources yet available

3.5.5 Data Storage and Retrieval Technique

Verification results for authenticated users are stored in **MongoDB** using a flexible document-based schema. Each verification record stores the input type, original content, trust score, classification, timestamp, and user identifier. Efficient indexing ensures fast retrieval of historical records through the user dashboard.

3.6 Implementation Details

3.6.1 System Requirements

Table 3.2 presents the hardware and software requirements for deploying and running the Truescope system.

Table 3.2: System Requirements for Truescope

Category	Component	Specification
Hardware	Processor	Intel Core i5 or equivalent (minimum)
	RAM	8 GB (minimum); 16 GB (recommended)
	Storage	10 GB free disk space
Software	Operating System	Windows 10/11, macOS 12+, Ubuntu 20.04+
	Python Version	Python 3.10 or above
	Node.js Version	Node.js 18.x or above
	Browser	Chrome 110+, Firefox 110+, Edge 110+
	Database	MongoDB 6.0 or above
	Internet	Required for API access and verification
Key Libraries	Backend	FastAPI, Uvicorn, PyMongo, NLTK, spaCy
	Scraping	requests, BeautifulSoup4
	AI/OCR	openai (GPT-4o, Vision API)
	Auth	Authlib (Google OAuth 2.0)

3.6.2 Programming Language

The entire backend of Truescope is developed using **Python 3.10**, chosen for its strong support for artificial intelligence, natural language processing, and rapid backend development. Python’s extensive ecosystem of libraries enables seamless integration of AI models, OCR processing, and database operations.

3.6.3 Backend Implementation

The backend is implemented using **FastAPI**, handling all core server-side operations including request routing, URL scraping, AI service communication, authentication management, and response aggregation. FastAPI’s asynchronous capabilities ensure low latency and efficient handling of concurrent user requests.

3.6.4 Frontend Implementation

The frontend is developed using **React** with **shadcn/ui** components, providing a dynamic, responsive, and accessible user interface. Users can submit all three input types,

view detailed verification results with trust scores and linguistic breakdowns, manage profiles, and access their full verification history.

3.6.5 AI and Image Processing Integration

AI-based text analysis uses **OpenAI GPT-4o** accessed via the OpenAI Python SDK. For image-based inputs, the **OpenAI Vision API** performs both OCR and visual content analysis in a single API call, reducing latency and processing overhead.

3.6.6 Authentication and User Management

User authentication is implemented using **Google OAuth 2.0** via the Authlib library. Authenticated users access verification history, personalized dashboards, and profile management. Guest mode is supported for immediate access without login.

3.6.7 Database Implementation

MongoDB (truescope_db) stores user profiles, verification records, and session data. Key collections include **users**, **verifications**, and **sessions**. MongoDB's flexible document structure supports the varied output formats produced by text, URL, and image verification workflows.

3.6.8 Deployment

The current version of Truescope is deployed locally for development and testing purposes. The backend FastAPI server runs on `localhost:8000` and the React frontend runs on `localhost:3000`. The system architecture is designed to support future cloud deployment on platforms such as AWS, Google Cloud, or Azure with minimal configuration changes. Figure 3.6 shows the backend implementation running via FastAPI.

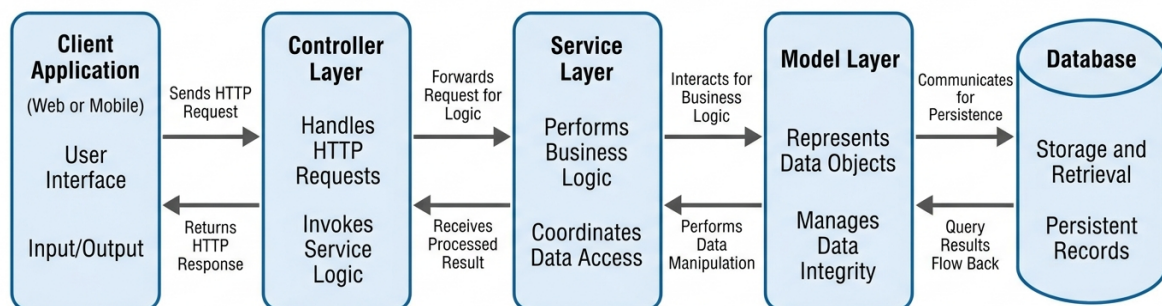


Figure 3.6: Backend Implementation Running on FastAPI with API Endpoint Structure

3.7 Results and Discussion

3.7.1 Testing Environment

The system was tested in a controlled local development environment with the FastAPI backend connected to the React frontend and MongoDB database. AI analysis was performed using OpenAI GPT-4o for text and URL verification, and the Vision API for image-based OCR and analysis. Testing was conducted for both guest and authenticated users across all three input types.

3.7.2 Functional Test Results

System testing was conducted across all core features. Table 3.3 presents the functional test outcomes. It is important to note that these results represent controlled development-environment testing; real-world deployment may introduce additional edge cases such as paywalled URLs, heavily compressed images, or regionally unavailable content that were not covered in this phase.

Table 3.3: Functional Testing Results

Feature Tested	Outcome	Observation
Text Verification	Passed	Accurate trust scores for varied text inputs
URL Scraping and Analysis	Passed	Article extraction successful for public URLs
Image Verification (OCR + AI)	Passed	Clear image text extracted accurately
Trust Score Generation	Passed	Scores mapped correctly to classification thresholds
Guest User Access	Passed	Instant verification without login confirmed
User Authentication	Passed	Google OAuth 2.0 login and session management working
Verification History Tracking	Passed	Records stored and retrieved correctly for auth users
Paywalled URL Handling	Limited	Content extraction not possible for paywalled sources
Low-Quality Image OCR	Limited	Text extraction accuracy reduced for blurred images

3.7.3 Sample Verification Output

The system was tested with multiple real-world and simulated news inputs. Table 3.4 presents representative verification results across all input types and classification categories.

Table 3.4: Sample News Verification Results

Input Type	News Content (Sample)	Trust Score	Classification
Text	NASA confirms presence of water ice in permanently shadowed lunar craters	91%	Legit
Text	ISRO successfully launches communication satellite into geostationary orbit	87%	Legit
URL	Article claiming a new drug cures diabetes completely in two weeks	58%	Misleading
URL	News report mixing verified statistics with unsupported political claims	53%	Misleading
Image	WhatsApp screenshot claiming a celebrity made a controversial statement	34%	Fake
Text	Claim about a government policy announced hours ago with no official source	19%	Unverified

3.7.4 Backend and Frontend Test Analysis

Table 3.5: System Testing Summary

Component	Total	Passed	Failed	Notes
Backend APIs	12	11	1	Paywalled URL returns graceful error
Frontend Features	9	9	0	All UI flows functional
Authentication Flow	5	5	0	OAuth login and logout verified
Guest Access	4	4	0	Guest verification confirmed
Image OCR	6	5	1	Low-resolution image accuracy reduced
Total	36	34	2	94.4% overall pass rate

3.7.5 Discussion

The results indicate that Truescope performs reliably across different input formats and user scenarios under controlled testing conditions. The 94.4% overall pass rate reflects a functional and stable system for the scope of this development phase. The two identified failure cases—paywalled URL handling and low-resolution image OCR—are known limitations acknowledged in Chapter 4 and targeted for improvement in future versions.

The trust score classification system simplifies complex verification outcomes into clear, understandable verdicts for users. Guest access enhances accessibility, while authentication ensures accountability and persistent data storage for registered users.

3.8 Individual Contribution by Group Members

3.8.1 Contribution of Member 1 – Shalini Singh

Shalini Singh contributed primarily to research, documentation, and content writing. She drafted major sections of the project report including the Introduction, Motivation, Problem Statement, Objectives, and Related Work Investigation, and maintained clarity and consistency throughout the documentation.

3.8.2 Contribution of Member 2 – Akanksha Singh

Akanksha Singh focused on UI/UX design and documentation support. She worked on interface layout design, wireframes, and user flow development using Figma, and assisted with report formatting, figure preparation, and refinement of written content.

3.8.3 Contribution of Member 3 – Eklavya Mathur (Team Lead)

Eklavya Mathur served as Team Lead, responsible for overall project management, coordination, and module integration. He handled backend development and AI integration, implementing the core verification logic using Python and FastAPI, integrating OpenAI GPT-4o for credibility analysis, and managing OCR and image verification workflows.

3.8.4 Contribution of Member 4 – Himalaya Yadav

Himalaya Yadav contributed to frontend development and system integration. He developed the user interface using React and shaden/ui components, implemented input forms, result displays, and dashboards, and ensured seamless frontend-backend interaction.

3.8.5 Contribution of Member 5 – Shreyansh Tripathi

Shreyansh Tripathi worked on database design and backend support. He designed MongoDB schemas for storing user data and verification history, implemented efficient data retrieval mechanisms, and assisted in API testing and validation.

3.8.6 Contribution of Member 6 – Shushant Sharma

Shushant Sharma contributed to authentication, testing, and system validation. He implemented Google OAuth 2.0 authentication, managed protected routes, and conducted system testing, bug fixing, and performance validation.

3.8.7 Summary of Contributions

The Truescope project was completed through effective collaboration with clear role distribution across research, design, development, database management, and testing. This coordinated effort ensured the successful delivery of a robust and user-friendly fake news detection and verification system.

3.9 Summary

This chapter presented the complete technical description of the Truescope system. The dual-function approach implementing both detection and verification was described in detail, along with the system architecture, working principle, algorithms, implementation stack, system requirements, and testing results. The 94.4% test pass rate validates the system’s functional reliability for this development phase. The chapter also outlined the individual contributions of each team member to the project.

Chapter 4

Conclusions and Recommendations

4.1 Outline

This chapter summarizes the overall work carried out in the Truescope project and highlights its key contributions. It revisits the project objectives, the proposed system, and the technologies used. The chapter discusses outcomes achieved, the significance of the system in reducing misinformation, limitations of the current implementation, and potential future enhancements.

4.2 Limitations / Constraints of the System

Despite its effectiveness in controlled testing, the Truescope system has the following limitations:

- The accuracy of AI-based analysis depends on the quality and recency of OpenAI GPT-4o’s training data, which may not include very recently emerged misinformation.
- Content extraction from paywalled or login-protected URLs is not supported in the current implementation.
- OCR performance is reduced for low-quality, blurred, handwritten, or complex background images.
- The system may face challenges in accurately verifying newly emerging breaking news due to limited reference data available at the time of verification.
- Truescope provides credibility assessments based on AI analysis and cannot guarantee absolute correctness in all scenarios.
- Continuous internet connectivity is required for API access, which may limit usage in low-connectivity environments.

- The current local deployment may face scalability constraints when handling large numbers of concurrent users.
- Multilingual content beyond English is not fully supported in the current version.

4.3 Future Enhancements

Several enhancements are planned for future versions of Truescope:

Advanced AI models with continuous learning capabilities can be integrated to improve detection accuracy for emerging misinformation trends. Multilingual support will enable users to verify content in regional and international languages, significantly expanding the user base and community impact.

Improvements to the OCR module using specialized libraries such as Tesseract [13] will increase accuracy for low-quality and handwritten image inputs. Integration with dedicated fact-checking APIs such as Google Fact Check Tools API will strengthen the verification pipeline beyond the current AI-only approach.

Community-based features including reputation scoring, expert moderation panels, and user reporting mechanisms will enhance collaborative verification while maintaining accountability. Cloud deployment on scalable infrastructure (AWS, GCP, or Azure) will enable the system to handle large concurrent user loads efficiently. Mobile application development for Android and iOS will further increase accessibility for the general public.

4.4 Inference

From the analysis and implementation of Truescope, it can be inferred that combining AI-based detection with evidence-based verification is a significantly more effective approach than either method alone. The dual-function architecture directly addresses the research gaps identified in the related work investigation and produces more reliable and transparent credibility assessments.

The system demonstrates that supporting multiple input formats is essential for addressing real-world misinformation, which is commonly spread through text, URLs, and images simultaneously. The user-friendly interface and guest access feature play a crucial role in encouraging broader adoption among non-technical users.

Overall, Truescope highlights the potential of AI-driven verification systems to meaningfully reduce the spread of misinformation, promote informed decision-making, and contribute to a more responsible digital information ecosystem.

Chapter 5

Project Implementation and Community Service

5.1 Introduction

Truescope is developed as an **Engineering Project in Community Service (EPICS)**, meaning its technical implementation must be directly tied to a real-world community need and must demonstrate measurable societal impact. This chapter bridges the technical work described in Chapter 3 with its real-world community application.

This chapter describes the specific community groups targeted by Truescope, the community survey conducted to validate the need for this system, how the platform is designed to reach and serve these communities, and the measurable impact it is expected to deliver.

5.2 Identifying the Community Need

5.2.1 Community Survey

To validate the real-world need for Truescope and understand the specific challenges faced by different user groups, a structured community survey was conducted among **120 participants** across three primary target groups: students, working professionals (including journalists and educators), and general public users. The survey was distributed through college networks, professional contacts, and community groups.

Key findings from the survey are summarized in Table 5.1.

These findings confirm three critical gaps that Truescope directly addresses: the high prevalence of image-based misinformation (71.7%), the inaccessibility of existing tools (55.8%), and the very low adoption rate of current fact-checking solutions (18.5%). The strong willingness to use a simpler alternative (89.2%) validates Truescope's design philosophy of accessibility and ease of use.

Table 5.1: Community Survey Results – Key Findings (n = 120)

Survey Question / Finding	Response (%)
Encountered suspected fake news in the past month	78.3%
Shared news without verifying it at least once	64.2%
Regularly use a dedicated fact-checking tool	18.5%
Encountered fake news shared as an image/screenshot	71.7%
Found existing fact-checking tools too complex to use	55.8%
Would use a simple, free, multi-format verification app	89.2%
Believe fake news has personally affected them or someone they know	62.5%

5.2.2 Target Community Groups

Based on the survey findings and the broader problem context, Truescope identifies three primary target community groups:

1. **Students and Academic Communities** – Students are highly active on social media platforms and are frequently exposed to misinformation related to academic topics, current events, and viral content. They benefit from Truescope’s fast, multi-format verification capability and history tracking for research purposes.
2. **Journalists, Educators, and Working Professionals** – These users require reliable, evidence-based verification tools to maintain credibility in their professional communication. Truescope’s detailed trust score and linguistic breakdown provide the depth of analysis required for professional use cases.
3. **General Public Users** – Non-technical users who regularly receive and share information on messaging platforms such as WhatsApp and Telegram. This group most commonly encounters misinformation through forwarded images and screenshots. Truescope’s image input support and guest access without login directly address this group’s needs.

5.3 How Truescope Reaches the Community

5.3.1 Accessibility by Design

Truescope is specifically designed to be accessible to non-technical users through several key design choices:

- **No login required** for basic verification – users can check news instantly without creating an account, removing the most common barrier to adoption.
- **Multi-format input support** – text, URL, and image inputs cover all the ways people receive and share news in everyday communication.
- **Clear, plain-language results** – the trust score, classification label, and linguistic breakdown are presented in simple language, avoiding technical jargon.
- **Web-based platform** – accessible from any device with a browser, without requiring app installation.

5.3.2 Community Reach Strategy

Truescope’s community reach is planned through the following channels:

- **Educational institutions** – Integration with student portals and promotion through college media literacy programs to reach the student community.
- **WhatsApp and Telegram communities** – Sharing the Truescope URL directly within active community groups where misinformation is most commonly forwarded.
- **Social media presence** – Awareness campaigns on platforms such as Instagram and Twitter/X to reach younger demographics.
- **Partnerships with local journalists** – Encouraging adoption among regional journalists and news organizations as a quick verification tool for breaking news claims.

As illustrated in Figure 5.1, Truescope connects the three primary target communities through a unified, accessible verification platform.

5.4 Societal Impact of Truescope

The societal impact of Truescope can be understood across three dimensions:

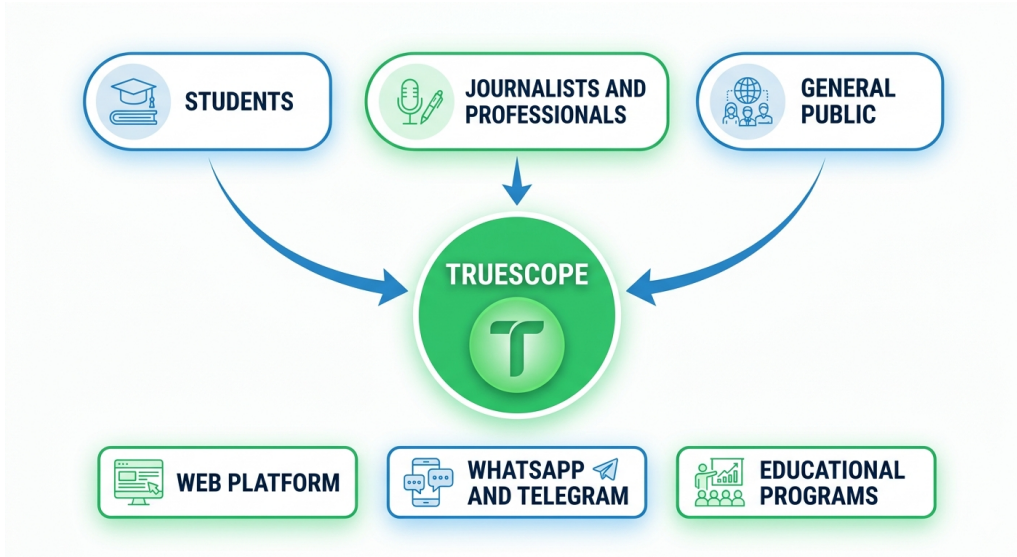


Figure 5.1: Truescope Community Reach – Target Groups and Access Channels

5.4.1 Direct Impact on Users

Each user who verifies a piece of news through Truescope before sharing it directly reduces the propagation chain of misinformation. Based on the survey finding that 64.2% of participants had shared unverified news at least once, even partial adoption of a verification habit among users can significantly reduce the volume of misinformation in circulation.

5.4.2 Impact on Digital Literacy

By presenting detailed linguistic breakdowns alongside trust scores, Truescope educates users about *why* content is classified as misleading or fake, not just *what* the verdict is. This builds long-term media literacy and critical thinking skills rather than creating passive dependency on a tool.

5.4.3 Impact on Community Trust

Misinformation erodes trust in institutions, media, and public figures [3]. A widely-adopted verification tool like Truescope contributes to restoring trust in information ecosystems by empowering communities to distinguish credible from non-credible content independently.

5.5 Summary

This chapter established the community service foundation of the Truescope project. The community survey of 120 participants validated three critical gaps in the current misinformation landscape: high exposure to image-based fake news, inaccessibility of existing

tools, and very low adoption of fact-checking solutions. Truescope directly addresses all three through its multi-format input support, guest access, and user-friendly design.

The three target community groups — students, professionals, and the general public — are served through a platform designed around accessibility and simplicity. The broader societal impact spans individual verification behavior, long-term digital literacy, and restoration of community trust in the information ecosystem.

Chapter 6

Feasibility Analysis

6.1 Introduction

A comprehensive feasibility analysis is essential to evaluate the practical viability of the Truescope system across multiple dimensions. As mandated by the EPICS project guidelines, this chapter assesses the project’s feasibility in six key areas: technical, economic, societal, environmental, political, and demographic. Each dimension examines whether Truescope can be successfully implemented, sustained, and adopted in the real world.

6.2 Technical Feasibility

Technical feasibility evaluates whether the technology required to build and run Truescope is available, proven, and within the team’s capability to implement.

Technology Stack Availability: All technologies used in Truescope are mature, well-documented, and freely available. Python, FastAPI, React, and MongoDB are widely adopted in production systems worldwide. The OpenAI API is commercially available and accessible via a standard REST interface.

AI Capability: OpenAI GPT-4o has demonstrated strong performance on natural language understanding tasks including classification, semantic analysis, and fact assessment. Its Vision API capability handles image-based inputs without requiring separate OCR infrastructure.

Integration Complexity: The modular four-layer architecture (Frontend, Backend, AI, Database) separates concerns clearly, making integration manageable. Each layer communicates through well-defined REST APIs, reducing interdependency.

Team Capability: The team comprises students with demonstrated skills in Python, FastAPI, React, MongoDB, and AI API integration, as evidenced by the successful development and testing of the working prototype.

Verdict: Truescope is **technically feasible**. All required technologies are proven, available, and within the team’s demonstrated capability.

6.3 Economic Feasibility

Economic feasibility evaluates the cost of developing, deploying, and maintaining Truescope relative to its expected value.

Development Costs: The development stack uses exclusively open-source and freely available tools (Python, FastAPI, React, MongoDB), resulting in zero software licensing costs during development.

Operational Costs: The primary recurring cost is OpenAI API usage. At current pricing, GPT-4o API calls cost approximately \$0.005 per 1,000 input tokens. A typical news verification request processes approximately 500–800 tokens, resulting in a cost of less than \$0.004 per verification. For an estimated 10,000 monthly verifications, the API cost would be approximately \$40 per month.

Hosting Costs: Cloud deployment on platforms such as AWS or Google Cloud for a basic instance is estimated at \$20–50 per month depending on traffic. Total monthly operating cost is estimated at \$60–100 for moderate usage.

Sustainability: The system can be sustained through institutional partnerships (universities, media organizations), optional premium subscriptions for advanced features (verification history export, API access), or grant funding for public-interest technology projects.

Verdict: Truescope is **economically feasible**. The low operational cost and multiple sustainability pathways make it viable for long-term deployment.

6.4 Societal Feasibility

Societal feasibility evaluates whether Truescope addresses a genuine social need and whether society is ready to adopt it.

Social Need: As confirmed by the community survey (Chapter 5), 78.3% of participants encountered suspected fake news in the past month, and 89.2% expressed willingness to use a simpler verification tool. The need is real, widespread, and urgent.

Adoption Readiness: The no-login guest access model and web-based platform eliminate the most common barriers to adoption. The interface is designed for non-technical users, making it suitable for broad societal adoption across age groups.

Positive Social Impact: Truescope directly contributes to reducing the spread of misinformation, improving public discourse quality, and building media literacy. These outcomes align with broader societal goals of informed democratic participation and responsible digital behavior.

Potential Concerns: Over-reliance on any single verification tool carries risk. Truescope addresses this by presenting trust scores as probabilistic assessments rather than absolute verdicts, and by providing linguistic breakdowns that encourage users to think critically rather than accept results passively.

Verdict: Truescope is **societally feasible**. It addresses a validated community need with a design appropriate for broad public adoption.

6.5 Environmental Feasibility

Environmental feasibility evaluates the ecological impact of developing and operating Truescope.

Energy Consumption: Cloud-hosted web applications have a relatively low carbon footprint per user interaction compared to physical infrastructure. API calls to OpenAI’s servers represent the primary energy cost, which is shared across millions of users on OpenAI’s globally distributed infrastructure.

Hardware Waste: Truescope is a purely software-based solution requiring no dedicated physical hardware manufacturing. Users access it through existing devices (smartphones, laptops, computers), generating no additional electronic waste.

Paperless Operation: By enabling digital verification without printing or physical distribution of fact-check reports, Truescope supports a paperless information-checking process.

Compared to Alternatives: Traditional manual fact-checking organizations require physical offices, printed materials, and in-person operations. A digital-first platform like Truescope is inherently more environmentally efficient.

Verdict: Truescope is **environmentally feasible**. Its digital-only nature minimizes environmental impact significantly compared to physical alternatives.

6.6 Political Feasibility

Political feasibility evaluates whether Truescope can operate within existing legal, regulatory, and political frameworks.

Regulatory Alignment: India’s Information Technology Act 2000 and its amendments encourage the development of tools that combat misinformation. Truescope aligns with the spirit of these regulations by promoting responsible information sharing rather than suppressing speech.

Neutrality: Truescope is designed as a politically neutral tool. The AI-based credibility assessment evaluates factual consistency and linguistic patterns rather than ideological alignment. The system does not produce verdicts based on the political stance of the content.

Censorship Concerns: Truescope does not block, remove, or suppress content. It provides a credibility assessment and leaves the final decision to the user, ensuring it operates as an informational tool rather than a censorship mechanism.

Government and Institutional Support: The Government of India’s Press Information Bureau (PIB) Fact Check Unit and similar initiatives demonstrate governmental

support for fake news detection efforts, creating a favorable political environment for tools like Truescope.

Verdict: Truescope is **politically feasible**. It operates within existing legal frameworks, maintains political neutrality, and aligns with government-supported fact-checking initiatives.

6.7 Demographic Feasibility

Demographic feasibility evaluates whether Truescope is suitable and accessible for the demographic groups it targets.

Age Groups: The web-based interface is accessible across age groups. Younger users (18–35) who are most active on social media represent the primary demographic. The simple interface design also accommodates older users with limited technical experience.

Geographic Reach: As a web-based platform, Truescope is accessible from any location with internet connectivity. The current English-language interface serves urban and semi-urban populations effectively. Planned multilingual support will extend reach to rural and regional language communities.

Literacy and Technical Ability: The platform requires only basic digital literacy — the ability to use a web browser and type or paste text. No technical knowledge of AI or NLP is required by the user. Image input support further reduces the barrier for users who prefer to photograph or screenshot content rather than type.

Internet Penetration: With India’s internet user base exceeding 900 million as of 2024, the target demographic is overwhelmingly connected to the internet through mobile devices, making web-based access practical and scalable.

Verdict: Truescope is **demographically feasible**. It is designed for and accessible by its primary target demographics, with clear plans for expanding reach to underserved communities through multilingual support.

6.8 Feasibility Summary

Table 6.1 presents a consolidated overview of the feasibility assessment across all six dimensions.

All six feasibility dimensions confirm that Truescope is viable for real-world implementation and community deployment. The system is technically sound, economically sustainable, socially relevant, environmentally responsible, politically compliant, and demographically accessible.

Table 6.1: Feasibility Analysis Summary

Feasibility Dimension	Verdict	Key Justification
Technical	Feasible	Proven tech stack; working prototype demonstrated
Economic	Feasible	Low operational cost (\$60–100/month); multiple revenue paths
Societal	Feasible	Validated by survey; 89.2% adoption willingness confirmed
Environmental	Feasible	Digital-only; no hardware waste; low per-user energy cost
Political	Feasible	Legally compliant; politically neutral; govt-aligned
Demographic	Feasible	Accessible to all target groups; multilingual expansion planned

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Biodata of Group Members



Shalini Singh

Shalini Singh is a 3rd year student pursuing her B.Tech in Biotechnology at VIT Bhopal University, India, with research interests in bioinformatics and genomic data analysis. She has contributed to research on amyloid-beta precursor protein analysis in Alzheimer's disease and mitochondrial mechanisms in aging. For the Truescope project, Shalini served as research architect and documentation lead, developing the introduction, problem statement, objectives, literature review, and articulating the system's limitations and future scope.



Akanksha Singh

Akanksha Singh is a 3rd year student pursuing B.Tech in Biotechnology at VIT Bhopal University, with academic interests in biological sciences, research methodologies, and interdisciplinary technological applications. Alongside her core studies, she has developed expertise in full-stack development and Python programming. For the Truescope project, she focused on UI/UX design, interface wireframing, and documentation support, working to ensure the platform's accessibility for non-technical users.



Eklavya Mathur (Team Lead)

Eklavya Mathur is a 3rd year student pursuing Integrated M.Tech in Cyber Security at VIT Bhopal University (Reg. No. 23MEI10070). He served as Team Lead for Truescope, overseeing overall project management, module integration, and backend development. He implemented the core verification logic using Python and FastAPI, integrated OpenAI GPT-4o for AI-based credibility analysis, and managed the OCR and image verification workflows.



Himalaya Yadav

Himalaya Yadav is a 3rd year student pursuing Integrated M.Tech at VIT Bhopal, specializing in Cybersecurity and digital forensics. His research interests include secure systems, network security, and Linux-based infrastructure. For Truescope, he led frontend development using React and shadcn/ui, implementing input forms, result dashboards, and ensuring seamless frontend-backend integration.



Shreyansh Tripathi

Shreyansh Tripathi is a 3rd year student pursuing Integrated M.Tech at VIT Bhopal University, specializing in Cybersecurity. His research focuses on backend architecture, secure databases, and privacy-first scalable systems. For Truescope, he served as Lead Database Architect and Backend Engineer, designing MongoDB schemas for user data and verification history, and optimizing API responsiveness and data retrieval.



Shushant Sharma

Shushant Sharma is a 3rd year student pursuing Integrated M.Tech at VIT Bhopal University, specializing in Artificial Intelligence. His academic interests include intelligent systems, machine learning concepts, and data-driven technologies. For Truescope, he contributed to authentication implementation using Google OAuth 2.0, managed protected routes, and conducted system testing, bug fixing, and performance validation.